Final Project
STAT 525

The final project in STAT 525 can be either a literature review or original research presentation.

- **Literature review.** The following is a list of related papers of interest, and this option should involve in-depth summaries and exposition of one of these papers. You can also present other papers related to the optimization, but permission from the instructor is required.

- **Original research.** You are encouraged to present your original research if your current research is related to optimization, which can be theoretic or experimental. If you choose this option, then you can do it individually or in groups of two and need to submit a proposal before Oct 15.

The deadline for choosing a paper or submitting a proposal is Oct 15. After that date, we will assign review papers for students who don't select. If your selected paper was assigned to other students, we will let you know and ask you to change your choice.

In the final project, you are required to submit a report and slides, and give a in-person presentation in the last two weeks:

1. You need to prepare a report for the selected paper or original research. The report should be prepared by LaTex.

2. You also need to prepare a 15 minutes presentation to summarize your report. After presentation, please submit your slides.

Here is the list of papers:

1. Lan, G., Li, Y. and Zhao, T., 2023. Block policy mirror descent. SIAM Journal on Optimization, 33(3), pp.2341-2378.

2. Zhang, Z. and Lan, G., 2024. Optimal methods for convex nested stochastic composite optimization. Mathematical Programming, pp.1-48.

3. Zhang, L., Yang, J., Karbasi, A. and He, N., 2024. Optimal guarantees for algorithmic reproducibility and gradient complexity in convex optimization. Advances in Neural Information Processing Systems, 36.

4. Zhao, R., Morwani, D., Brandfonbrener, D., Vyas, N. and Kakade, S., 2024. Deconstructing What Makes a Good Optimizer for Language Models. arXiv preprint arXiv:2407.07972.

5. Vyas, N., Morwani, D., Zhao, R., Kaplun, G., Kakade, S. and Barak, B., 2023. Beyond implicit bias: The insignificance of sgd noise in online learning. arXiv preprint arXiv:2306.08590.

6. Vyas, N., Morwani, D., Zhao, R., Shapira, I., Brandfonbrener, D., Janson, L. and Kakade, S., 2024. SOAP: Improving and Stabilizing Shampoo using Adam. arXiv preprint arXiv:2409.11321.

7. Robbins, H. and Monro, S., 1951. A stochastic approximation method. The annals of mathematical statistics, pp.400-407.

8. Bottou, L. and Bousquet, O., 2007. The tradeoffs of large scale learning. Advances in neural information processing systems, 20.

9. Su, W., Boyd, S. and Candes, E.J., 2016. A differential equation for modeling Nesterov's accelerated gradient method: Theory and insights. The Journal of Machine Learning Research, 17(1), pp.5312-5354.

10. Moulines, E. and Bach, F., 2011. Non-asymptotic analysis of stochastic approximation algorithms for machine learning. Advances in neural information processing systems, 24.

11. Dieuleveut, A., Durmus, A. and Bach, F., 2020. Bridging the gap between constant step size stochastic gradient descent and markov chains.

12. Varre, A.V., Pillaud-Vivien, L. and Flammarion, N., 2021. Last iterate convergence of SGD for Least-Squares in the Interpolation regime. Advances in Neural Information Processing Systems, 34, pp.21581-21591.

13. Li, Q., Tai, C. and Weinan, E., 2019. Stochastic modified equations and dynamics of stochastic gradient algorithms i: Mathematical foundations. The Journal of Machine Learning Research, 20(1), pp.1474-1520.

14. Zhang, C., Bengio, S., Hardt, M., Recht, B. and Vinyals, O., 2021. Understanding deep learning (still) requires rethinking generalization. Communications of the ACM, 64(3), pp.107-115.

15. Keskar, N.S., Mudigere, D., Nocedal, J., Smelyanskiy, M. and Tang, P.T.P., 2016. On large-batch training for deep learning: Generalization gap and sharp minima. arXiv preprint arXiv:1609.04836.

16. Vaswani, S., Bach, F. and Schmidt, M., 2019, April. Fast and faster convergence of sgd for over-parameterized models and an accelerated perceptron. In The 22nd international conference on artificial intelligence and statistics (pp. 1195-1204). PMLR.

17. Ma, S., Bassily, R. and Belkin, M., 2018, July. The power of interpolation: Understanding the effectiveness of SGD in modern over-parametrized learning. In International Conference on Machine Learning (pp. 3325-3334). PMLR.

18. Pesme, S., Pillaud-Vivien, L. and Flammarion, N., 2021. Implicit bias of sgd for diagonal linear networks: a provable benefit of stochasticity. Advances in Neural Information Processing Systems, 34, pp.29218-29230.

19. Vivien, L.P., Reygner, J. and Flammarion, N., 2022, June. Label noise (stochastic) gradient descent implicitly solves the lasso for quadratic parametrisation. In Conference on Learning Theory (pp. 2127-2159). PMLR.

20. Ali, A., Dobriban, E. and Tibshirani, R., 2020, November. The implicit regularization of stochastic gradient flow for least squares. In International conference on machine

learning (pp. 233-244). PMLR.

21. Soudry, D., Hoffer, E., Nacson, M.S., Gunasekar, S. and Srebro, N., 2018. The implicit bias of gradient descent on separable data. The Journal of Machine Learning Research, 19(1), pp.2822-2878.

22. Chizat, L. and Bach, F., 2020, July. Implicit bias of gradient descent for wide two-layer neural networks trained with the logistic loss. In Conference on Learning Theory (pp. 1305-1338). PMLR.

23. Vaskevicius, T., Kanade, V. and Rebeschini, P., 2019. Implicit regularization for optimal sparse recovery. Advances in Neural Information Processing Systems, 32.

24. Woodworth, B., Gunasekar, S., Lee, J.D., Moroshko, E., Savarese, P., Golan, I., Soudry, D. and Srebro, N., 2020, July. Kernel and rich regimes in overparametrized models. In Conference on Learning Theory (pp. 3635-3673). PMLR.

25. Gunasekar, S., Lee, J., Soudry, D. and Srebro, N., 2018, July. Characterizing implicit bias in terms of optimization geometry. In International Conference on Machine Learning (pp. 1832-1841). PMLR.

26. Gunasekar, S., Woodworth, B. and Srebro, N., 2021, March. Mirrorless mirror descent: A natural derivation of mirror descent. In International Conference on Artificial Intelligence and Statistics (pp. 2305-2313). PMLR.

27. Li, Z., Wang, T., Lee, J.D. and Arora, S., 2022. Implicit bias of gradient descent on reparametrized models: On equivalence to mirror descent. Advances in Neural Information Processing Systems, 35, pp.34626-34640.

28. Gunasekar, S., Woodworth, B.E., Bhojanapalli, S., Neyshabur, B. and Srebro, N., 2017. Implicit regularization in matrix factorization. Advances in neural information processing systems, 30.

29. Bartlett, P.L., Long, P.M., Lugosi, G. and Tsigler, A., 2020. Benign overfitting in linear regression. Proceedings of the National Academy of Sciences, 117(48), pp.30063-30070.

30. Belkin, M., Rakhlin, A. and Tsybakov, A.B., 2019, April. Does data interpolation contradict statistical optimality?. In The 22nd International Conference on Artificial Intelligence and Statistics (pp. 1611-1619). PMLR.

31. Ji, Z. and Telgarsky, M., 2019, June. The implicit bias of gradient descent on nonseparable data. In Conference on Learning Theory (pp. 1772-1798). PMLR.

32. Ali, A., Kolter, J.Z. and Tibshirani, R.J., 2019, April. A continuous-time view of early stopping for least squares regression. In The 22nd international conference on artificial intelligence and statistics (pp. 1370-1378). PMLR.

33. Suggala, A., Prasad, A. and Ravikumar, P.K., 2018. Connecting optimization and regularization paths. Advances in Neural Information Processing Systems, 31.

34. Martin, C.H. and Mahoney, M.W., 2021. Implicit Self-Regularization in Deep Neural Networks: Evidence from Random Matrix Theory and Implications for Learning. J. Mach. Learn. Res., 22(165), pp.1-73.

35. Lan, G. and Zhou, Y., 2018. An optimal randomized incremental gradient method. Mathematical programming, 171(1), pp.167-215.

36. Charisopoulos, V., Chen, Y., Davis, D., Díaz, M., Ding, L. and Drusvyatskiy, D., 2021. Low-rank matrix recovery with composite optimization: good conditioning and rapid convergence. Foundations of Computational Mathematics, pp.1-89.

37. Hong, M., Luo, Z.Q. and Razaviyayn, M., 2016. Convergence analysis of alternating direction method of multipliers for a family of nonconvex problems. SIAM Journal on Optimization, 26(1), pp.337-364.

38. Arora, S., Cohen, N. and Hazan, E., 2018, July. On the optimization of deep networks: Implicit acceleration by overparameterization. In International Conference on Machine Learning (pp. 244-253). PMLR.

39. Mei, S., Bai, Y. and Montanari, A., 2016. The landscape of empirical risk for nonconvex losses. arXiv preprint arXiv:1607.06534.

40. Chi, Y., Lu, Y.M. and Chen, Y., 2019. Nonconvex optimization meets low-rank matrix factorization: An overview. IEEE Transactions on Signal Processing, 67(20), pp.5239-5269.

41. Carmon, Y., Duchi, J.C., Hinder, O. and Sidford, A., 2018. Accelerated methods for nonconvex optimization. SIAM Journal on Optimization, 28(2), pp.1751-1772.

42. Du, S., Lee, J., Li, H., Wang, L. and Zhai, X., 2019, May. Gradient descent finds global minima of deep neural networks. In International conference on machine learning (pp. 1675-1685). PMLR.

43. Mu, C., Zhang, Y., Wright, J. and Goldfarb, D., 2016. Scalable robust matrix recovery: Frank–Wolfe meets proximal methods. SIAM Journal on Scientific Computing, 38(5), pp.A3291-A3317.

44. Sun, J., Qu, Q. and Wright, J., 2018. A geometric analysis of phase retrieval. Foundations of Computational Mathematics, 18(5), pp.1131-1198.

45. Sun, J., Qu, Q. and Wright, J., 2016. Complete dictionary recovery over the sphere I: Overview and the geometric picture. IEEE Transactions on Information Theory, 63(2), pp.853-884.

46. Sun, J., Qu, Q. and Wright, J., 2016. Complete dictionary recovery over the sphere II: Recovery by Riemannian trust-region method. IEEE Transactions on Information Theory, 63(2), pp.885-914.

47. Ge, R., Lee, J.D. and Ma, T., 2016. Matrix completion has no spurious local minimum. Advances in neural information processing systems, 29.